

Employee ID Card Request System using ServiceNow

Capstone Project Documentation

1. Introduction

Project Title

Employee ID Card Request System using ServiceNow

2. Project Overview

Purpose

Organizations frequently receive requests from employees for new or replacement ID cards due to loss, damage, expiry, or changes in employee information. In many organizations, these requests are handled manually through emails, paper forms, or verbal communication, resulting in delays, poor request tracking, and increased administrative workload.

The purpose of this project is to automate the Employee ID Card Request process using the ServiceNow platform. The application enables employees to submit requests through a Service Catalog item while automating approval, email notification, and task creation. By eliminating manual intervention, the system provides a faster, more transparent, and efficient request management process.

Features

The Employee ID Card Request System provides the following features:

- Employee ID Card Request through Service Catalog
 - Employee Name, Department, and Reason as input fields
 - Automated Approval Workflow using Flow Designer
 - Approval Notification through Email
 - Automatic Catalog Task Creation
 - Request Tracking using ServiceNow records
 - Centralized Request Management
 - Improved workflow automation
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Project Description

The Employee ID Card Request System is a workflow automation solution developed on the ServiceNow platform. The primary objective of the system is to digitize the employee ID card request process and eliminate the delays associated with manual request handling.

Employees can access the Service Catalog and submit a request by providing their Employee Name, Department, and the reason for requesting a new or replacement ID card. Once the request is submitted, Flow Designer automatically retrieves the submitted information and initiates the approval workflow.

The approval request is routed to the designated approver. During development and testing, the approval was assigned to the System Administrator user. After approval, the workflow automatically sends an email notification informing the employee that the request has been approved.

Following the successful approval, the workflow creates a Catalog Task for the administration team. This task represents the operational activity required to print and issue the employee ID card. Since every request generates its own task, administrators can easily monitor pending work and process requests efficiently.

The system demonstrates how ServiceNow can automate business processes using Service Catalog, Flow Designer, Approval Management, Notifications, and Catalog Tasks while maintaining complete traceability of every request.

Benefits of the System

The proposed solution provides several advantages over manual request processing.

- Eliminates paper-based request handling.
 - Reduces approval delays.
 - Improves transparency in request processing.
 - Provides automated notifications.
 - Automatically assigns work to the administration team.
 - Maintains centralized records.
 - Improves employee satisfaction.
 - Demonstrates workflow automation using ServiceNow.
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3. Solution Architecture

The Employee ID Card Request System follows a layered architecture to separate user interaction, workflow automation, approval processing, and data management. Each layer is responsible for a specific functionality, making the application easier to maintain, monitor, and extend in the future.

The architecture begins with employees interacting with the Service Catalog interface. After submitting a request, Flow Designer performs the business logic by retrieving the catalog variables, sending the request for approval, notifying the employee upon approval, and finally generating a

Catalog Task for the administration team.

All requests, approvals, tasks, and notifications are stored within ServiceNow tables, ensuring complete traceability and centralized management.

3.1 Presentation Layer (Frontend)

The Presentation Layer is implemented using the ServiceNow Service Catalog.

Employees interact with the application through the **Employee ID Card Request** Catalog Item.

The catalog form contains the following variables:

- Employee Name
- Department
- Reason

This interface allows employees to submit requests quickly without requiring technical knowledge. The Service Catalog provides a clean, user-friendly environment where all required information is captured before submission.

3.2 Business Logic Layer

The Business Logic Layer is implemented using **Flow Designer**.

Once a request is submitted, the Flow Designer automatically executes the workflow.

The workflow performs the following operations:

- Retrieves Employee Name.
- Retrieves Department.
- Retrieves Reason.
- Sends the request for approval.
- Waits for approval decision.
- Checks whether the request is approved.
- Sends an approval email.
- Creates a Catalog Task.

Because all these operations are automated, the need for manual intervention is significantly reduced.

3.3 Approval Layer

The Approval Layer manages the authorization process before an ID card can be issued.

After submission, the request is automatically forwarded to the designated approver. For this project, the approval was configured to the **System Administrator** account for testing purposes.

The approver can review the submitted request and either approve or reject it. Only approved requests proceed to the next stage of the workflow.

This ensures that unauthorized ID card requests are prevented while maintaining organizational control over the issuance process.

3.4 Notification Layer

The Notification Layer improves communication between employees and administrators.

Once the approval is completed, Flow Designer automatically sends an email notification to the employee.

The email contains confirmation that the request has been approved and informs the employee that further processing will be carried out by the administration team.

Automatic notifications eliminate the need for employees to repeatedly check the status of their requests.

3.5 Task Management Layer

The Task Management Layer is responsible for creating operational work for the administration team.

After approval, Flow Designer automatically generates a **Catalog Task (sc_task)**.

The task represents the work required to print, prepare, and issue the employee ID card.

Using Catalog Tasks enables administrators to:

- Track pending work.
 - Manage task completion.
 - Monitor request fulfillment.
 - Improve accountability.
 - Reduce processing delays.
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3.6 Data Layer

The Data Layer stores all application data inside the ServiceNow platform.

The major tables involved include:

- **sc_req_item** – Stores Requested Item records.
- **sc_task** – Stores Catalog Tasks.
- **sysapproval_approver** – Stores approval information.
- **sys_email** – Stores email notification records.

These tables maintain complete information about submitted requests, approvals, generated tasks, and notification history.

The centralized database allows administrators to retrieve records easily for auditing, reporting, and future reference.

4. Setup Instructions

The Employee ID Card Request System was developed and configured on the ServiceNow platform. The application uses Service Catalog and Flow Designer to automate the complete lifecycle of an employee ID card request. The following steps describe the setup procedure followed during project implementation.

Prerequisites

Before implementing the project, the following prerequisites are required:

- ServiceNow Developer Instance (Zurich Release)
 - Administrator access to the instance
 - Service Catalog Plugin
 - Flow Designer Plugin
 - Email Notification functionality enabled
 - Basic knowledge of ServiceNow administration
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Step 1: Create the Service Catalog Item

Navigate to:

Service Catalog → Catalog Definitions → Maintain Items

Create a new Catalog Item named **Employee ID Card Request**.

Configure the necessary details such as:

- Name
- Catalog
- Category
- Short Description
- Availability

This catalog item serves as the entry point for employees to submit their ID card requests.

Step 2: Create Catalog Variables

Add the required variables to the catalog item.

The variables created are:

- Employee Name
- Department
- Reason

These variables capture the necessary information required to process an employee's request.

Step 3: Configure Flow Designer

Navigate to:

Process Automation → Flow Designer

Create a new flow named:

Employee ID Card Request Flow

Choose **Service Catalog** as the trigger so that the workflow starts automatically whenever an employee submits the request.

Step 4: Configure Approval Workflow

Add the **Ask For Approval** action to the flow.

For testing purposes, configure the approval to be assigned to the **System Administrator** user.

This ensures every submitted request undergoes an approval process before further processing.

Step 5: Configure Email Notification

After approval, add the **Send Email** action.

Configure the email with an appropriate subject such as:

"Your Employee ID Card Request has been approved."

The email is automatically sent to the employee after successful approval.

Step 6: Create Catalog Task

Add the **Create Catalog Task** action after the approval step.

The task is automatically generated for the administration team, indicating that the employee's ID card is ready to be printed and issued.

Step 7: Test the Application

Submit a sample Employee ID Card Request.

Verify the following:

- Request submission
- Approval generation
- Approval completion
- Email notification
- Catalog Task creation
- Request completion

Successful verification confirms that the workflow is functioning correctly.

5. Folder / Component Structure

The Employee ID Card Request System is organized into multiple logical components within the ServiceNow platform. Each component performs a specific function in the overall workflow.

1. Service Catalog Layer

This layer contains the Employee ID Card Request Catalog Item.

It provides the interface through which employees submit their requests.

The layer includes:

- Employee Name
- Department
- Reason

The Service Catalog simplifies request submission while ensuring standardized data collection.

2. Catalog Variables

The Catalog Variables define the information collected from employees.

The project contains three variables:

- Employee Name
- Department
- Reason

These variables are retrieved later by Flow Designer for workflow execution.

3. Flow Designer Layer

Flow Designer manages the complete workflow automation.

The flow contains the following actions:

- Trigger
- Get Catalog Variables
- Ask For Approval
- Decision (Approved)
- Send Email
- Create Catalog Task

This component automates the request without requiring manual intervention.

4. Approval Management

This module controls the approval process.

The request is forwarded to the configured approver.

The approver can:

- Approve Request
- Reject Requests

5. Notification Module

This module automatically informs employees after approval.

Email notifications improve communication and reduce manual follow-up.

6. Task Management Module

The Catalog Task module creates operational tasks for the administration team.

Administrators use these tasks to:

- Print ID Cards
 - Issue ID Cards
 - Update Task Status
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7. Database Layer

The database layer stores all application records.

Major ServiceNow tables include:

- sc_req_item
- sc_task
- sysapproval_approver
- sys_email

These tables maintain complete request history for future reference.

Overall Component Structure

The project follows a modular architecture consisting of:

- User Interface Layer
- Service Catalog Layer
- Workflow Automation Layer
- Approval Layer
- Notification Layer
- Task Management Layer
- Database Layer

This modular approach improves maintainability, scalability, and ease of future enhancements.

6. Running the Application

The Employee ID Card Request System can be executed directly through the ServiceNow platform after configuration. The workflow is completely automated from request submission to task creation.

Step 1: Access the Service Catalog

The employee logs into the ServiceNow instance using valid credentials.

From the Service Catalog, the employee selects the **Employee ID Card Request** catalog item.

Step 2: Fill the Request Form

The employee enters the required information:

- Employee Name
- Department
- Reason

After verifying the entered details, the employee submits the request.

Step 3: Request Creation

After submission, ServiceNow automatically creates:

- Request (REQ)
- Requested Item (RITM)

The request is stored in the ServiceNow database.

Step 4: Workflow Execution

Flow Designer is automatically triggered.

The workflow performs the following operations:

- Retrieves Catalog Variables
- Sends Request for Approval

- Waits for Approval
- Evaluates Approval Status

No manual intervention is required during this stage.

Step 5: Approval Processing

The System Administrator reviews the request.

If approved:

- Approval status changes to Approved.
- Workflow proceeds automatically.

If rejected:

- Workflow terminates without creating further tasks.
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Step 6: Email Notification

Immediately after approval, the system sends an email notification to the employee.

The employee receives confirmation that the request has been approved.

Step 7: Catalog Task Generation

Flow Designer automatically creates a Catalog Task.

The task is assigned for processing by the administration team.

The generated task remains open until the ID card is prepared and issued.

Step 8: Request Completion

After task completion, the request lifecycle is completed.

All related records remain available within ServiceNow for tracking, auditing, and future reference.

7. Authentication

The Employee ID Card Request System utilizes the built-in authentication and authorization mechanisms provided by the ServiceNow platform to ensure secure access to the application. Only authenticated users can access the Service Catalog and submit ID card requests. The platform validates user credentials before granting access, ensuring that unauthorized users cannot interact with the system.

7.1 Authentication Mechanism

The system relies on ServiceNow's native login process. Every user must sign in using valid ServiceNow credentials before accessing the Employee ID Card Request Catalog Item. User authentication, session management, and identity verification are handled automatically by the ServiceNow platform, eliminating the need for custom authentication logic.

7.2 Role-Based Access Control (RBAC)

ServiceNow provides Role-Based Access Control (RBAC) to restrict access to different parts of the application.

Typical roles include:

Employee

- Access the Service Catalog.
- Submit Employee ID Card Requests.
- View their submitted requests.
- Receive email notifications.

System Administrator

- Configure Catalog Items.
- Modify Flow Designer workflows.
- Approve or reject requests (used during testing).
- Monitor requests and generated tasks.

7.3 Data Security

The system ensures that request information remains secure by utilizing ServiceNow's built-in security features.

Security measures include:

- User authentication before accessing the application.
- Secure storage of request records.
- Controlled access through user roles.
- Protected approval workflow.
- Server-side workflow execution.

These mechanisms ensure that only authorized users can submit, approve, or manage Employee ID Card Requests.

7.4 Request Security

Each submitted request is stored securely within the ServiceNow database. Request records contain employee information, approval details, email history, and catalog task information. Since workflow execution occurs on the server side, users cannot bypass approval or manipulate the workflow.

8. User Interface

The Employee ID Card Request System provides a simple and user-friendly interface developed using the ServiceNow Service Catalog. The interface enables employees to submit requests easily while providing administrators with an efficient workflow for processing those requests.

8.1 Employee Request Form

The primary interface is the Employee ID Card Request Catalog Item.

Employees enter the following details:

- Employee Name
- Department
- Reason

The catalog form provides a clean layout that allows employees to complete the request within a few seconds.

8.2 Request Submission

After entering the required information, employees submit the request through the Service Catalog.

Upon successful submission:

- A Request (REQ) is created.
- A Requested Item (RITM) is generated.
- Flow Designer is triggered automatically.

This provides a seamless experience without requiring manual intervention.

8.3 Approval Interface

The approver reviews the submitted request through the ServiceNow Approval module.

The approver can:

- View Employee Details.
- Verify Department.
- Review the Reason.
- Approve or Reject the Request.

This interface ensures that only authorized requests proceed for further processing.

8.4 Email Notification

Once the request is approved, the employee receives an automated email notification.

The notification includes:

- Approval Confirmation.
- Request Information.
- Next Processing Status.

Automatic communication improves transparency and keeps employees informed throughout the request lifecycle.

8.5 Catalog Task Interface

Following approval, ServiceNow automatically generates a Catalog Task.

The administration team can:

- View Task Details.
- Track Pending Tasks.
- Update Task Status.
- Complete the Request.

This interface enables efficient fulfillment of ID card requests.

8.6 User Experience

The application has been designed with the following principles:

- Simple navigation.
- Minimal user input.
- Automated workflow.
- Clear approval process.
- Faster request processing.
- Easy task management.
- Improved usability.

The interface ensures that both employees and administrators can perform their tasks with minimal effort.

9. Testing

The Employee ID Card Request System was tested thoroughly to ensure that every module functioned according to the project requirements. Manual testing was performed to validate request submission, workflow execution, approval processing, email notifications, and catalog task generation.

9.1 Test Scope

Testing covered the following modules:

- Employee ID Card Request Form
- Catalog Variables
- Flow Designer Workflow
- Approval Process
- Email Notification
- Catalog Task Creation
- End-to-End Workflow

9.2 Test Cases

Test Case 1: Employee Request Submission

Input:

Employee enters Name, Department, and Reason.

Expected Result:

Request is submitted successfully.

Result:

Pass

Test Case 2: Request Record Generation

Input:

Employee submits the catalog form.

Expected Result:

REQ and RITM records are created.

Result:

Pass

Test Case 3: Approval Workflow

Input:

Submitted Request.

Expected Result:

Approval request is generated for the configured approver.

Result:

Pass

Test Case 4: Approval Decision

Input:

Approver selects Approve.

Expected Result:

Approval state changes to Approved.

Result:

Pass

Test Case 5: Email Notification

Input:

Request Approved.

Expected Result:

Approval confirmation email is sent.

Result:

Pass

Test Case 6: Catalog Task Creation

Input:

Approval Completed.

Expected Result:

Catalog Task (SCTASK) is generated automatically.

Result:

Pass

Test Case 7: End-to-End Workflow

Input:

Complete request lifecycle.

Expected Result:

Request, Approval, Email, and Catalog Task execute successfully.

Result:

Pass

10. Known Issues

The Employee ID Card Request System was successfully implemented and tested. However, a few limitations remain that can be addressed in future versions.

1. Single Approver Configuration

For testing purposes, approval is assigned only to the System Administrator. In a production environment, approvals should be dynamically assigned to the employee's reporting manager.

2. Limited Request Information

The current version collects only Employee Name, Department, and Reason. Additional fields such as Employee ID, Contact Number, or Photo Upload can be included in future releases.

3. No Status Dashboard

The project does not currently provide a dashboard for employees to monitor request progress. Future versions can include personalized request tracking dashboards.

4. No Mobile Optimization

The project was developed primarily for desktop usage. A responsive mobile interface could improve accessibility for employees.

5. Limited Reporting

The application stores all request information but does not currently provide graphical reports or analytics for administrators.

11. Future Enhancements

The Employee ID Card Request System can be expanded significantly to improve functionality and organizational value.

1. Manager-Based Approval

Automatically determine the employee's reporting manager and route requests for approval without manual configuration.

2. Employee ID Validation

Validate employee information against organizational records before accepting requests.

3. File Attachment Support

Allow employees to upload photographs or supporting documents for ID card processing.

4. Request Status Tracking

Provide employees with a dashboard to monitor the progress of their requests in real time.

5. SMS and Mobile Notifications

In addition to email notifications, integrate SMS or mobile push notifications to keep employees informed about request updates.

6. Reporting Dashboard

Develop dashboards showing:

- Total Requests
- Approved Requests
- Pending Requests
- Rejected Requests
- Average Processing Time

7. Multi-Level Approval Workflow

Support multiple approval stages involving Team Leads, Department Managers, HR, and Administration before issuing an ID card.

8. Admin Dashboard

Create a dedicated dashboard where administrators can manage pending catalog tasks, view completed requests, and monitor overall system performance.

Conclusion

The Employee ID Card Request System successfully demonstrates how ServiceNow can automate a common organizational process using Service Catalog and Flow Designer. The project streamlines the complete request lifecycle by enabling employees to submit requests, routing them for approval, sending automated email notifications, and creating Catalog Tasks for the administration team.

The implementation reduces manual effort, improves request visibility, and enhances operational efficiency while maintaining centralized records within the ServiceNow platform. The project serves as a practical example of workflow automation and showcases the core capabilities of ServiceNow in developing real-world business applications.